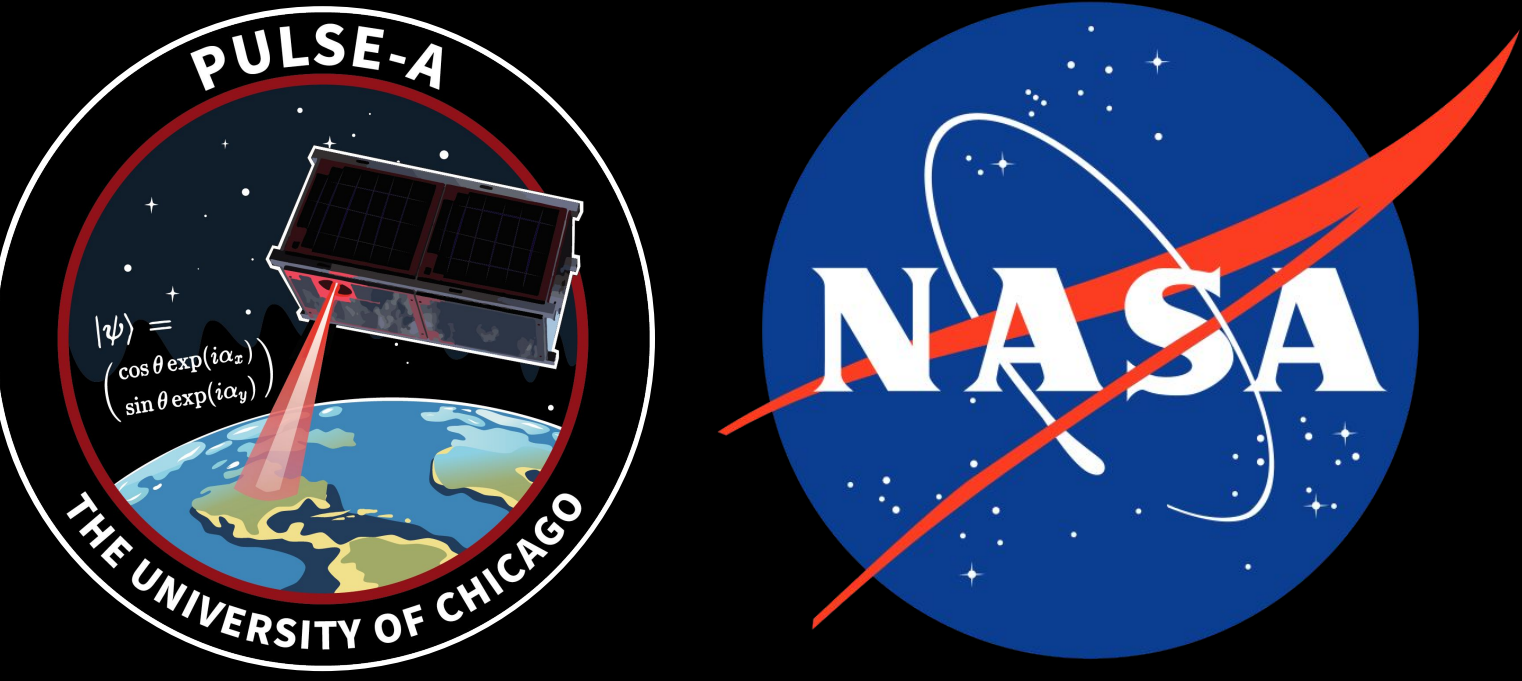




FINE POINTING CORRECTION FOR THE PULSE-A CUBESAT PAYLOAD

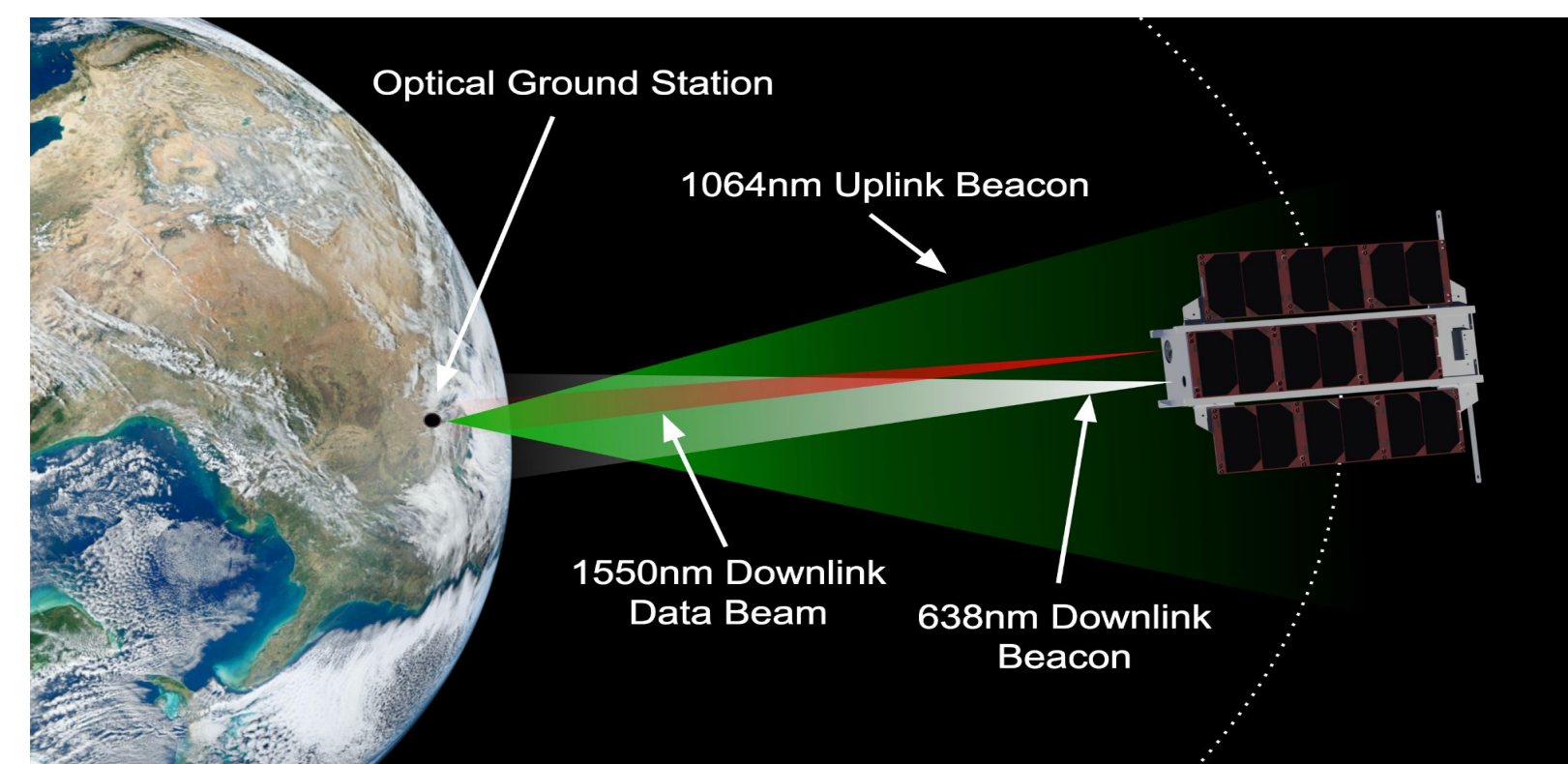
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PULSE-A MISSION

PULSE-A is a 3U (10cm x 10cm x 34cm) satellite designed by UChicago undergraduates, sponsored for launch by NASA's CubeSat Launch Initiative (CSLI). PULSE-A's purpose is to demonstrate space to ground optical communications via circular polarization shift keying.



MEMS FAST STEERING MIRROR

For fine PAT, PULSE-A will use a Mirrorcle MEMS FSM (Micro-electro-mechanical system fast steering mirror) to steer the uplink beacon and downlink data beam to within 0.7 degrees of pointing error. In order to provide sufficient fine pointing correction, simulations indicate at least 4.1 deg of pointing correction must be provided, with at least a 5 mm mirror diameter to catch the beam. Current payload designs include a Mirrorcle bonded FSM, with ± 4.21 deg mechanical rotation about both axes and a 6.4mm diameter. MEMS FSMs use the BQD (Biased Quadrant Differential) driving principle, where a bias voltage is applied across the actuators of both x and y axes. Differential voltages are then applied to cause angular displacements around a certain axis. The voltage response of a MEMS FSM can be modeled as a damped, driven harmonic oscillator, with a high quality factor corresponding to a fast FSM settling time. The BQD driving principle and bonded nature of the mirror make it highly repeatable and high-resolution, with at least 14 bits (16384 positions) on each axis and driving capabilities past 100 Hz.

$$I \frac{d^2 \theta}{dt^2} + \gamma \frac{d\theta}{dt} + k\theta = \tau(t)$$

$$H(s) = \frac{1/I}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

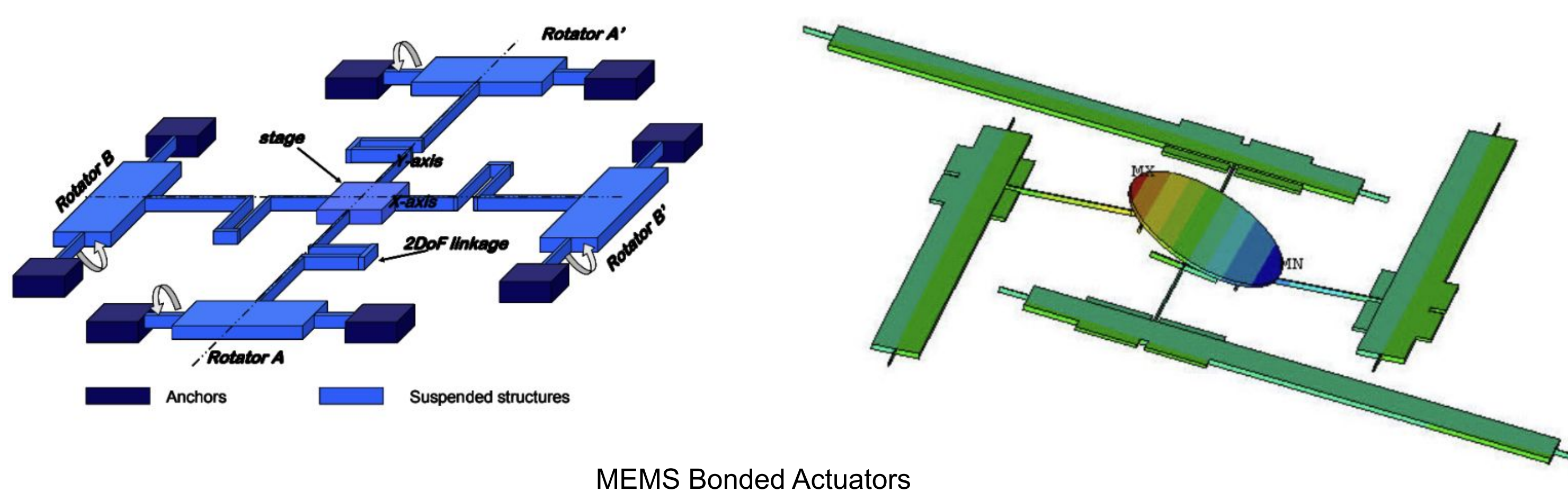
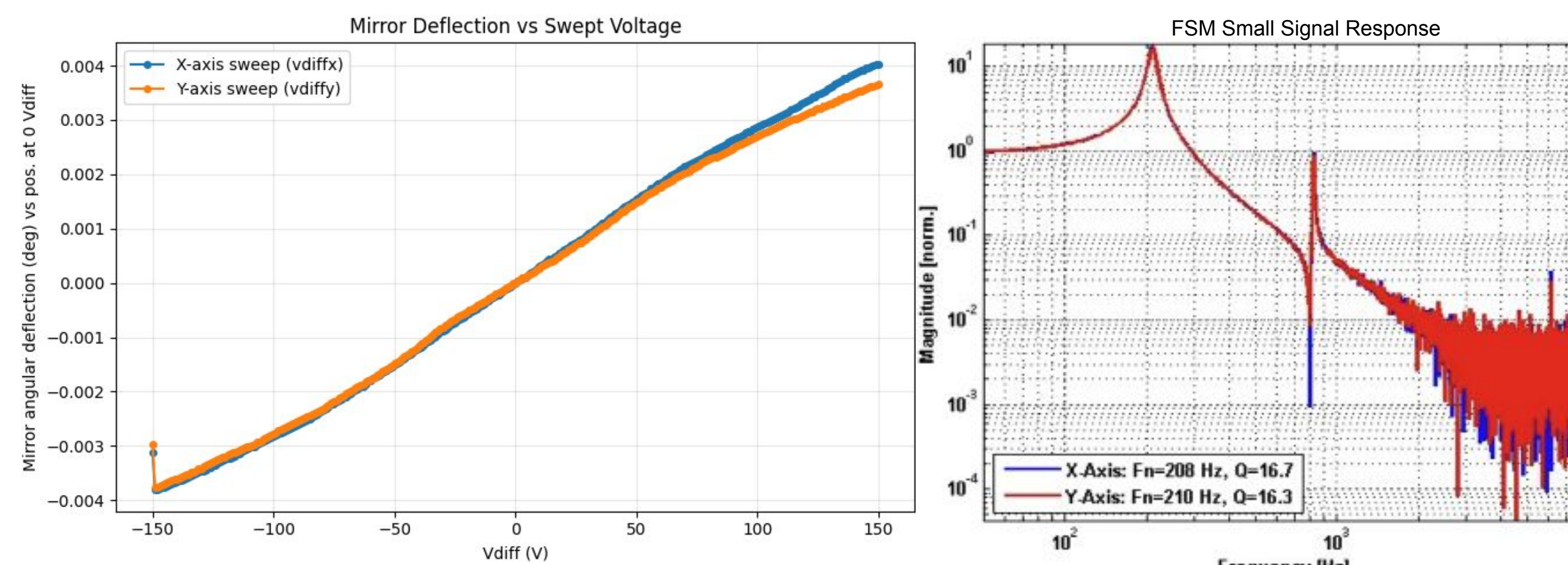
$$V_A = V_{bias} + \Delta V_x$$

$$V_C = V_{bias} - \Delta V_x$$

$$V_B = V_{bias} + \Delta V_y$$

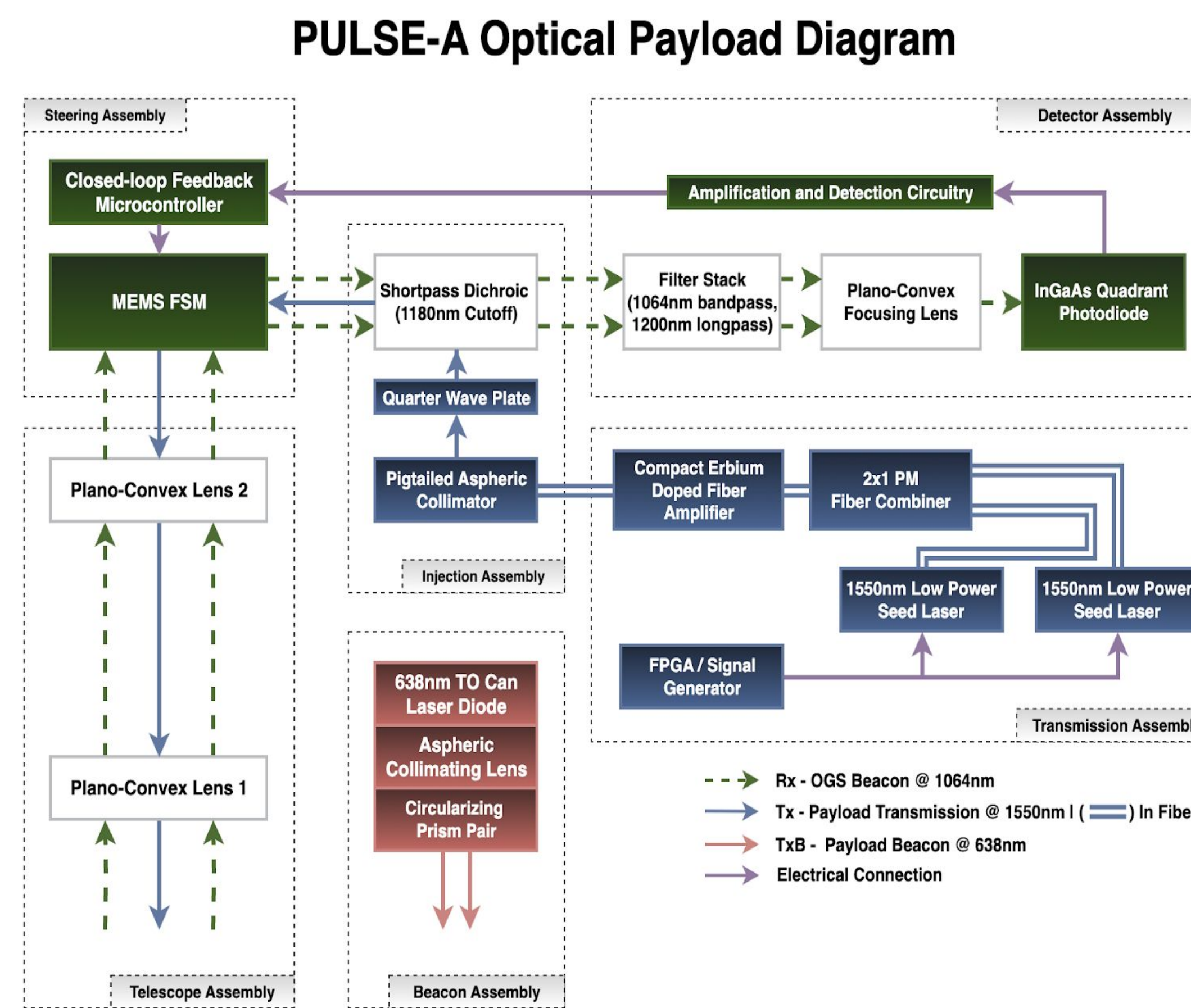
$$V_D = V_{bias} - \Delta V_y$$

$$Q = \frac{1}{2\zeta}$$



OPTICAL PAYLOAD

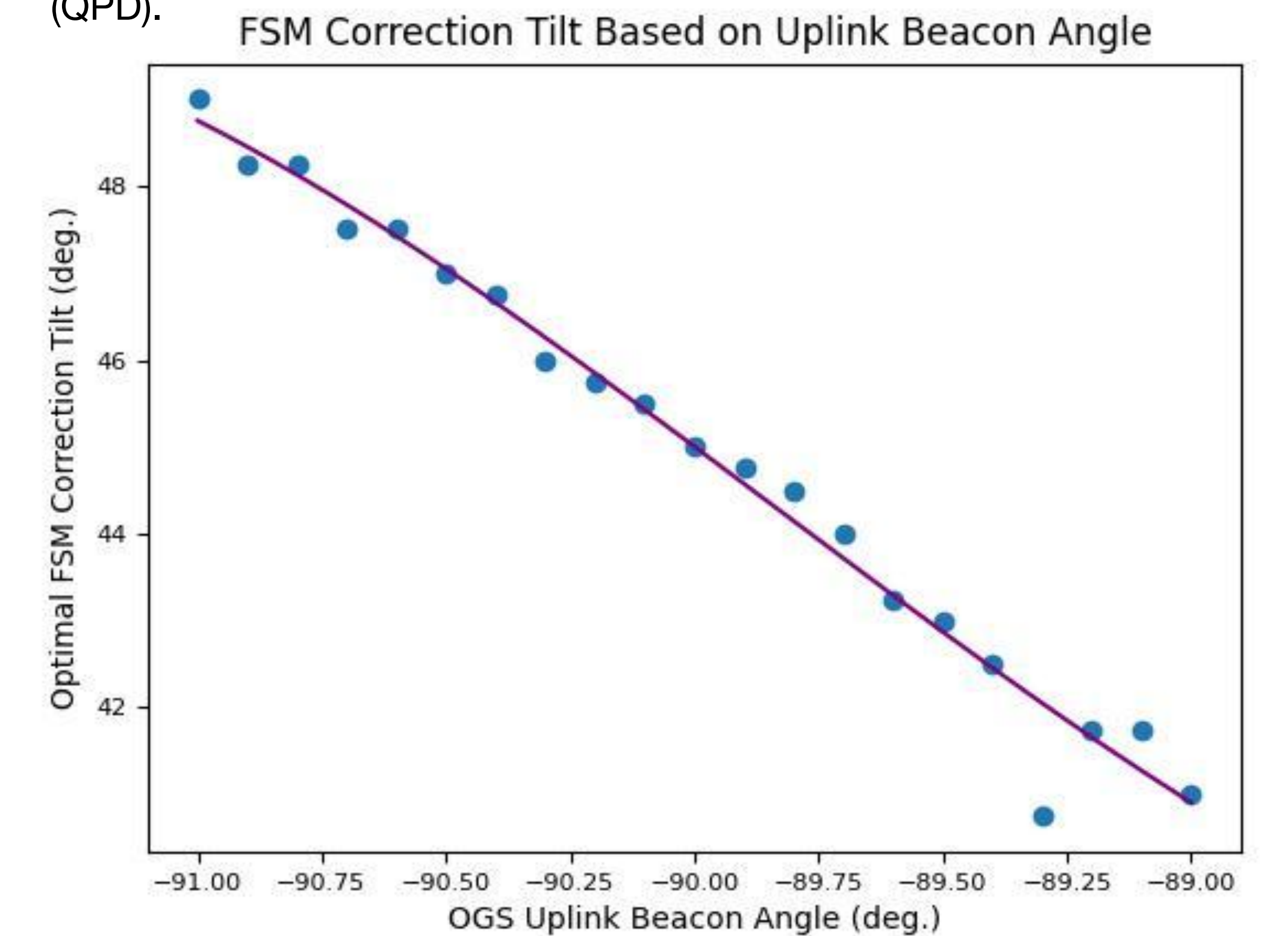
The PULSE-A payload uses three lasers at 638 nm, 1064 nm, and 1550 nm to establish contact and transmit to the OGS. The transmission assembly prepares a 1550 nm beam in two orthogonal, highly pure linear polarization states ($DOP \geq 95\%$) which is then amplified via an EDFA, boosting the optical power from 5 mW to 250 mW. The transmission beam is then collimated and converted to RHCP and LHCP by a quarter wave plate in the injection assembly. This beam passes through the steering assembly, which uses a closed-loop feedback mechanism between a quadrant photo diode (QPD) and the FSM to maintain the OGS beacon laser on a QPD setpoint. Both the incoming and outgoing signals (OGS 1064 nm beacon and 1550 nm payload transmission) are along the same optical path and controlled by a 1180 nm cutoff short-pass dichroic mirror. The beacon assembly transmits a collimated 638 nm laser for detection by the OGS.



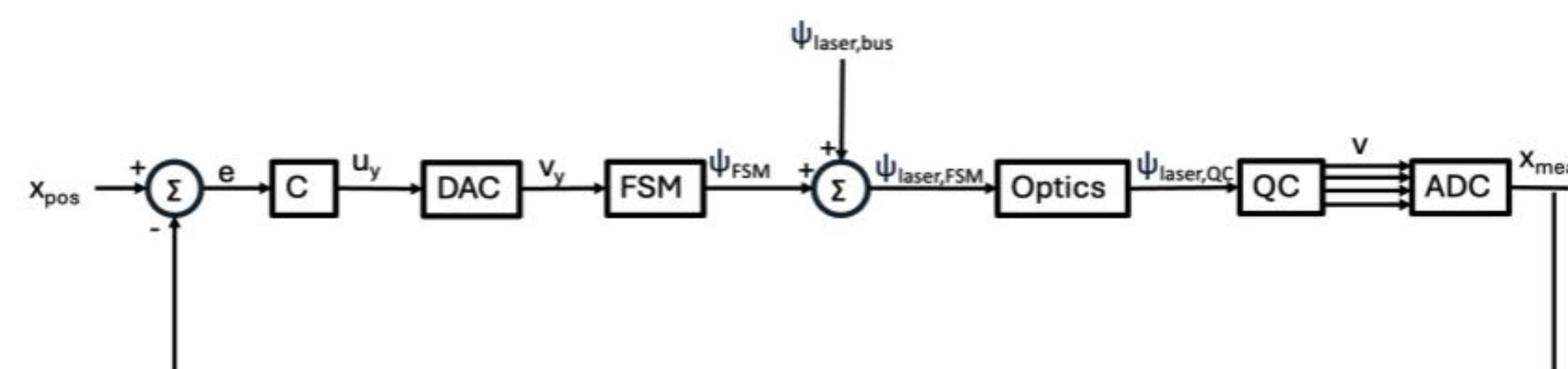
Mansilla, Sofia et al. "SSC25-P1-04 Development of a Compact Optical Communications Terminal for the PULSE-A CubeSat."

FINE POINTING

PULSE-A's altitude determination and control (ADC) system can steer the satellite to within 1 degree of pointing error, which is not enough for the optical payload's required 0.7 degrees of pointing error. Simulations demonstrate that the satellite will need ~8 deg of total correction tilt in order to maintain transmission alignment, which will be provided by an FSM in closed loop with a quad photo diode (QPD).



CLOSED LOOP CONTROL FOR FINE POINTING

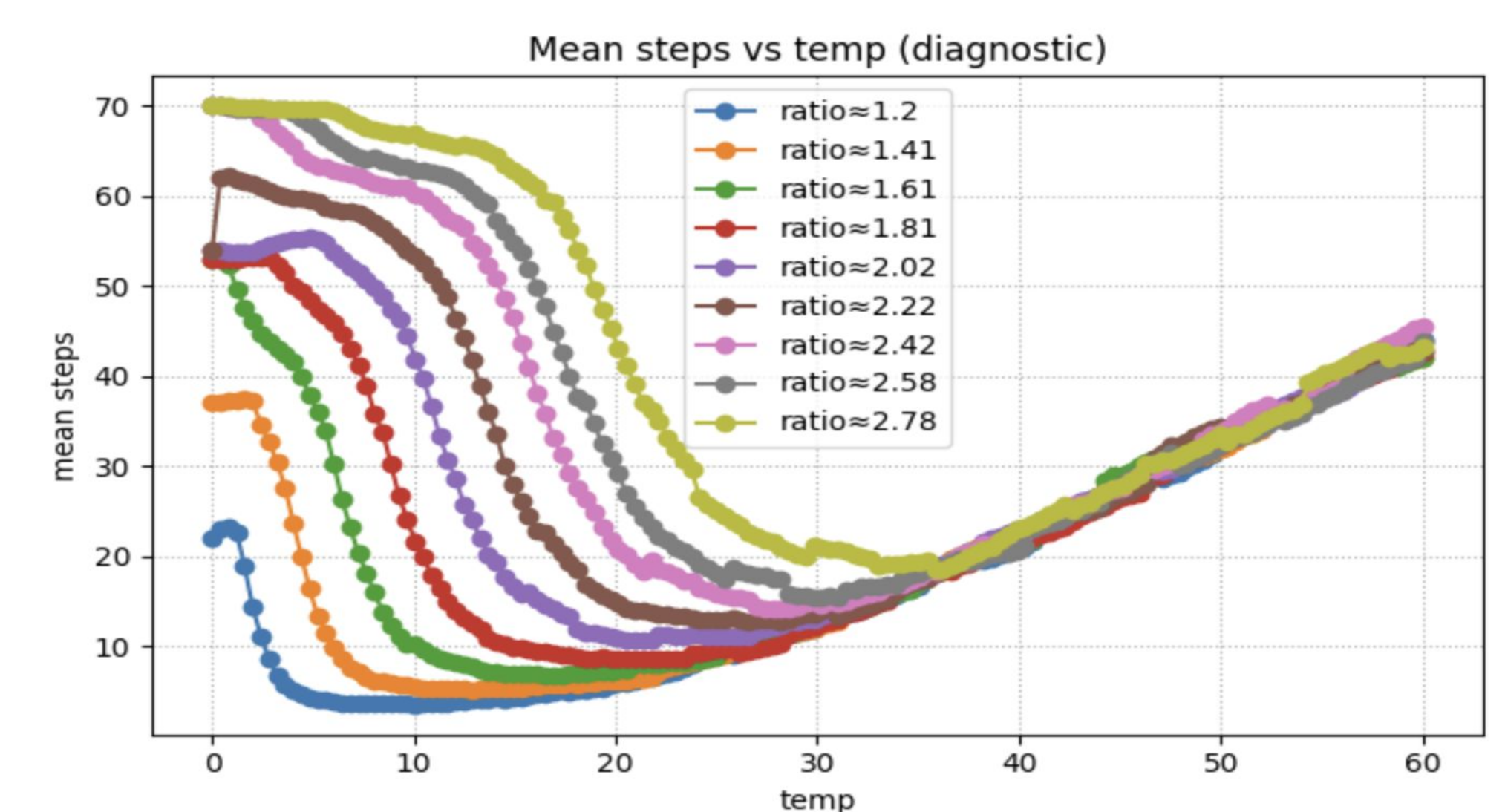


The steering feedback loop between the FSM and QPD operates by digitizing the voltage distribution from the QPD and processing it with an algorithm running on the Payload controller. The controller then outputs analog voltages to actuate the fast steering mirror (FSM), steering the beam back to the QPD center and thus performing fine pointing of the transmission beam. This closed-loop pointing algorithm serves to center the OGS beacon on the QPD by producing an equal voltage distribution across all four quadrants. The algorithm itself is based roughly on those previously developed and described by Rödiger in [5], and generally works to estimate the centroid of the incident beam using the relative intensities. The QPD refresh rate is anticipated to be the limiting factor for driving the feedback loop, and consequently the FSM driving rate will be limited to on the order of 10-50 Hz, well within its quasi-static regime. Current algorithm implementation involves a position, derivative, and integral control (PID) loop.

To first position the OGS beacon on the QPD, the FSM will execute an open-loop spiral-search protocol to enter closed-loop control within 10 seconds of the transmission window. Current simulations utilize a standard log spiral as given below:

$$\begin{bmatrix} x_{n+1} \\ y_{n+1} \end{bmatrix} = e^{b\Delta\theta} \begin{bmatrix} \cos(\Delta\theta) & -\sin(\Delta\theta) \\ \sin(\Delta\theta) & \cos(\Delta\theta) \end{bmatrix} \begin{bmatrix} x_n \\ y_n \end{bmatrix}$$

with random displacements injected into the spiral search, as a randomized spiral search algorithm converges to the QPD within the fewest iterations for certain search area geometries. Simulated search patterns give a 98% convergence rate with on the order of 200 iterations per convergence, QPD position dependent. With a conservative step time of <5 ms, this places search convergence and transition to closed loop control on the order of seconds.



References

- [1] Mansilla, Sofia et al. "SSC25-P1-04 Development of a Compact Optical Communications Terminal for the PULSE-A CubeSat."
- [2] Forester, Paige. Precision Pointing for the CubeSat Laser Infrared Crosslink (CLICK) Mission. Massachusetts Institute of Technology. Department of Mechanical Engineering, Feb. 2025.
- [3] Mirrorcle Technologies inc. *Mirrorcle MEMS Mirrors - Technical Overview*. 2021.
- [4] X. Zhao, Y. Yao, Y. Sun, and C. Liu, "Circle Polarization Shift Keying With Direct Detection for Free-Space Optical Communication," *J. Opt. Commun. Netw.*, vol. 1, no. 4, p. 307, 2009, doi: 10.1364/JOCN.1.000307.
- [5] A. Carrasco-Casado et al., "LEO-to-ground optical communications using SOTA (Small Optical Transponder) - Payload verification results and experiments on space quantum communications," *Acta Astronautica*, vol. 139, pp. 377-384, 2017, doi: 10.1016/j.actaastro.2017.07.030.
- [6] T. Ritz, J. Conklin, D. Mayer, J. Stupl, J. Hanson, and C. Engineering, "Testing of the CubeSat Laser Infrared Crosslink (CLICK-A) Payload."